Smart Event Management

Infosys | Springboard

AGENTIC AI · EVENT MANAGEMENT PLATFORM

AGENTIC AI POWERED

Smart Event Management System



A full-stack platform with an autonomous LangGraph agent and hybrid RAG for event discovery, registration and administration.

PRESENTED BY – TEAM C

Swati, Nithin, Pushpa, Sonali, Yugant, Sravanthi, Sushmitha & Sumit

UNDER THE GUIDANCE OF

Mr. Thangarasu M

Infosys | Springboard

Introduction

An agentic AI platform to create, discover, register for and administer events. An autonomous LangGraph agent reasons over each request, calls live tools and answers from policy documents.

Attendees
Search, register, pay, give feedback

Admins
Manage events, venues, users, payments

AI Agent
Role-aware tools, policy-grounded answers

12 role-aware AI tools

9 REST API groups

13 database tables

EventBot Hi! I search events, register you and answer policy questions.

Problem Statement & Objectives

THE PROBLEM

Scattered tools

Listings, registrations, venues and payments in separate tools

Manual tracking

Admins track seats and bookings by hand

No smart help

Chatbots answer but cannot act on live data

WITH THE AGENT

- One REST API for all event workflows
- JWT login with USER, ADMIN and MAIN_ADMIN roles
- LangGraph agent with 12 live, role-aware tools
- Hybrid RAG for grounded policy answers

EventBot I replace scattered tools and manual tracking with one smart assistant.

Infosys | Springboard

Tools & Technology Stack

Frameworks

- Python 3.11, FastAPI, Uvicorn
- React 19 + Vite, React Router
- Three.js 3D visuals
- Pydantic validation

LLM, AI & Retrieval

- LangGraph agent (LangChain)
- gpt-oss-120b via OpenRouter
- bge-small-en-v1.5 embeddings
- MarkItDown + hybrid RAG

Data & Storage

- PostgreSQL + pgvector
- SQLAlchemy ORM, Alembic
- Full-text search (tsvector)
- Redis cache

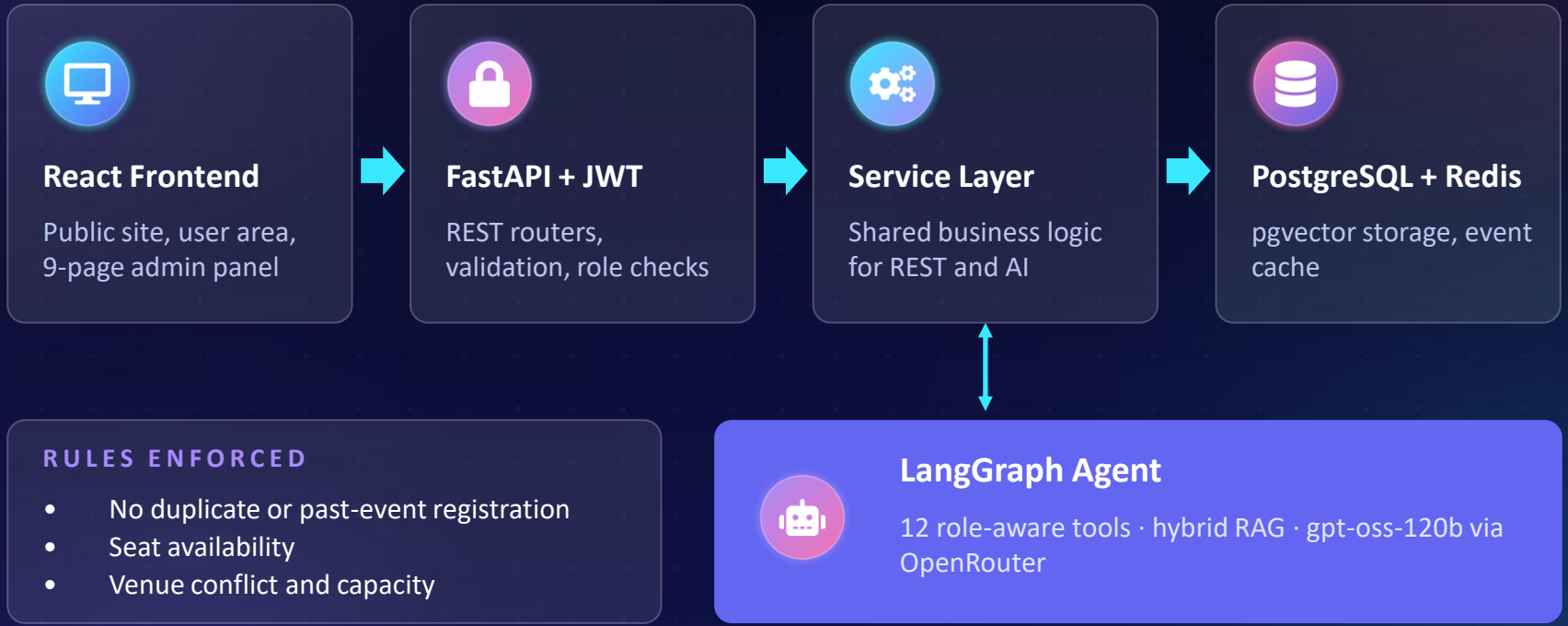
Security & Deployment

- JWT + bcrypt, role-based access
- pytest, 166 tests passing
- Docker Compose, uv, npm
- Git and GitHub

EventBot Under my hood: React, FastAPI, PostgreSQL + pgvector and LangGraph.

Infosys | Springboard

System Architecture



EventBot Every tool I use goes through the same service layer, never raw SQL.

Infosys | Springboard

Features by Role

For Attendees

- AI-driven event search
- Real-time seat tracking
- Duplicate and past-event checks
- Payment and feedback
- AI chat assistant

For Admins (ADMIN / MAIN_ADMIN)

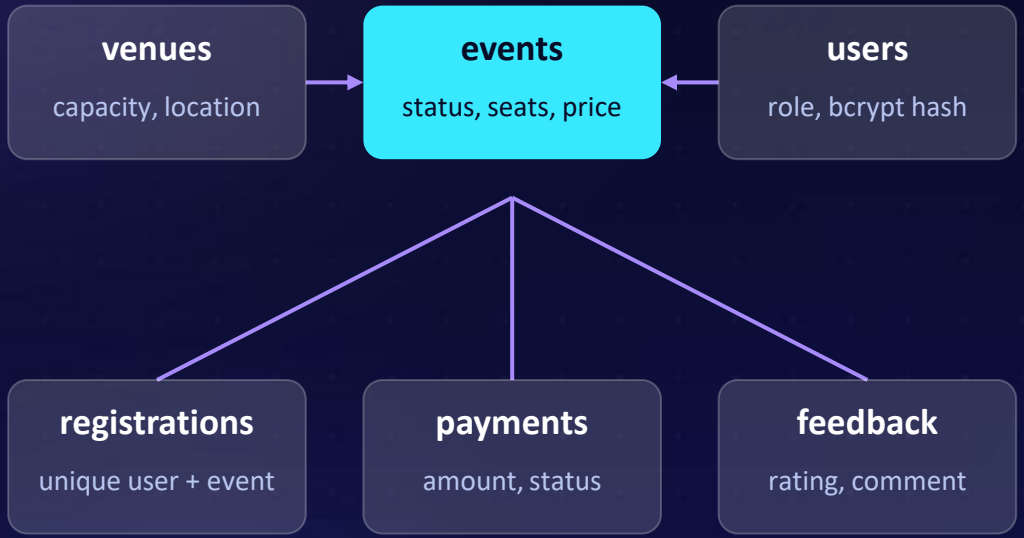
Dashboard	Events	Registrations
Users	Venues	Feedback
Payments	AI Assistant	Agent Logs

Role-checked actions • Every AI run and tool call is logged

EventBot Attendees chat with me; admins also get a 9-module control panel.

Infosys | Springboard

Database Design



Registrations, payments and feedback link to users and events by foreign keys.

13 tables on PostgreSQL + pgvector, 9 Alembic migrations

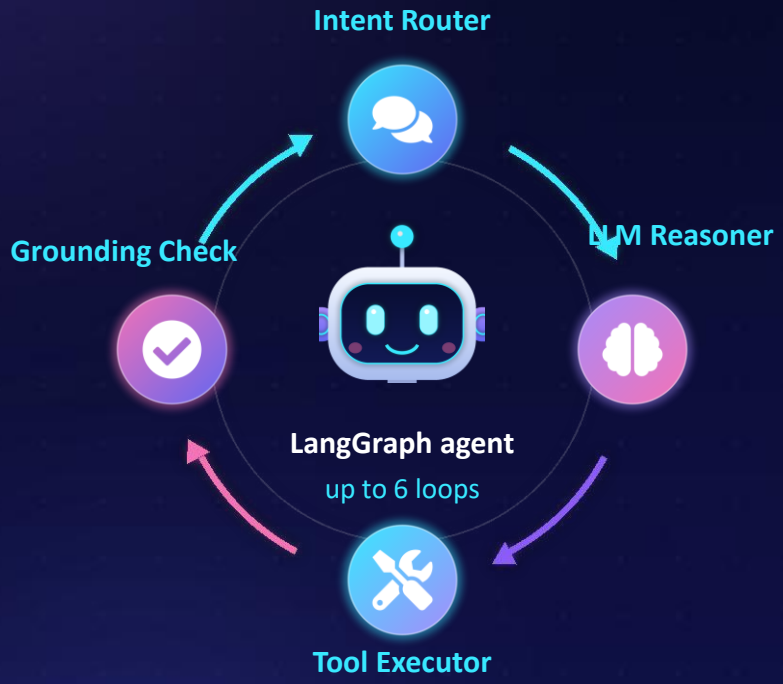
AGENT & KNOWLEDGE TABLES

- agent_sessions, chat_messages
- agent_runs, tool_calls (audit trail)
- knowledge_documents
- knowledge_chunks: 384-d vectors, HNSW + tsvector GIN
- Staged 2048-d chunk table with rollback

EventBot My memory lives here, including every run and tool call I make.

Infosys | Springboard

Agent Architecture



01 Intent Router

JWT sets identity and role; picks 1 of 11 intents; loads last 10 messages

02 LLM Reasoner

gpt-oss-120b via OpenRouter, with fallback models, picks a tool or answers

03 Tool Executor

12 role-aware tools reuse REST services; RAG search for policy questions

04 Grounding Check

Retries once or refuses; every run is logged

EventBot My pipeline: intent router, LLM reasoner, tool executor, grounding check.

Infosys | Springboard

Hybrid RAG & Grounded Answers



01

Ingest

MarkItDown reads policy docs; small chunks merge into neighbours



02

Index

384-d bge-small vectors with HNSW + tsvector GIN



03

Search

Vector + keyword search, merged by RRF



04

Rerank

Reranked down to the top 4 chunks



05

Answer

LLM answers; grounding check retries once or refuses

4 chunks after rerank

2048-d staged index with rollback

1 retry, then refuse



EventBot I answer from policy documents, and refuse if I cannot ground it.

Infosys | Springboard

Challenges & Solutions

Silent retrieval bugs Short policy sections dropped	→ ✓	Small chunks merge into neighbours
Changing embedding models Embedding size changed	→ ✓	2048-d index staged beside 384-d, with rollback
Agent inventing IDs Invented record IDs	→ ✓	Strict prompts, venue lookup tool, identity from JWT
Slow free models Replies took 40 to 50 s	→ ✓	Fallback models, retries and feature flags

EventBot Every bug taught me something. Here is how we fixed them.

Infosys | Springboard

Future Scope

Notifications & real payments

Email/SMS alerts and a live payment gateway

01

Recommendations & analytics

Interest-based suggestions and insight dashboards

02

Speed & scale

Faster model, background ingestion, 2048-d index

03

Cloud deployment

Hosted Docker, HTTPS, rate limiting, monitoring

04

EventBot Next: alerts, real payments, recommendations and cloud hosting.

Infosys | Springboard

Conclusion

One platform for events, registrations, venues, payments and feedback, plus a LangGraph agent with 12 role-aware tools and hybrid RAG for grounded answers.



Full-stack
13 tables · 9 API groups

Agentic
12 tools · hybrid RAG

Reliable
166 tests passing

Extensible
Payments, alerts, cloud

Thank You!